

Anomaly Detection & Dimensionality Reduction

Isolation Forest, t-SNE, UMAP & PCA — applied to jet engine anomaly detection and silicon thermal hotspot discovery.

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PROJECT 1

Jet Engine Anomaly Detection — Detect sensor anomalies in jet engine telemetry using Isolation Forest with t-SNE/UMAP visualization

1 Overview

Objective: Detect anomalous sensor readings from jet engine telemetry data. The dataset contains 4,650 recordings from 8 sensors with a 3.2% anomaly rate. Isolation Forest identifies anomalies by their isolation depth in random trees, while t-SNE and UMAP provide powerful 2D projections for visual analysis.

Domain: Aerospace & Predictive Maintenance

Anomaly Types: Turbine overheat, Oil pressure loss, Fan blade imbalance

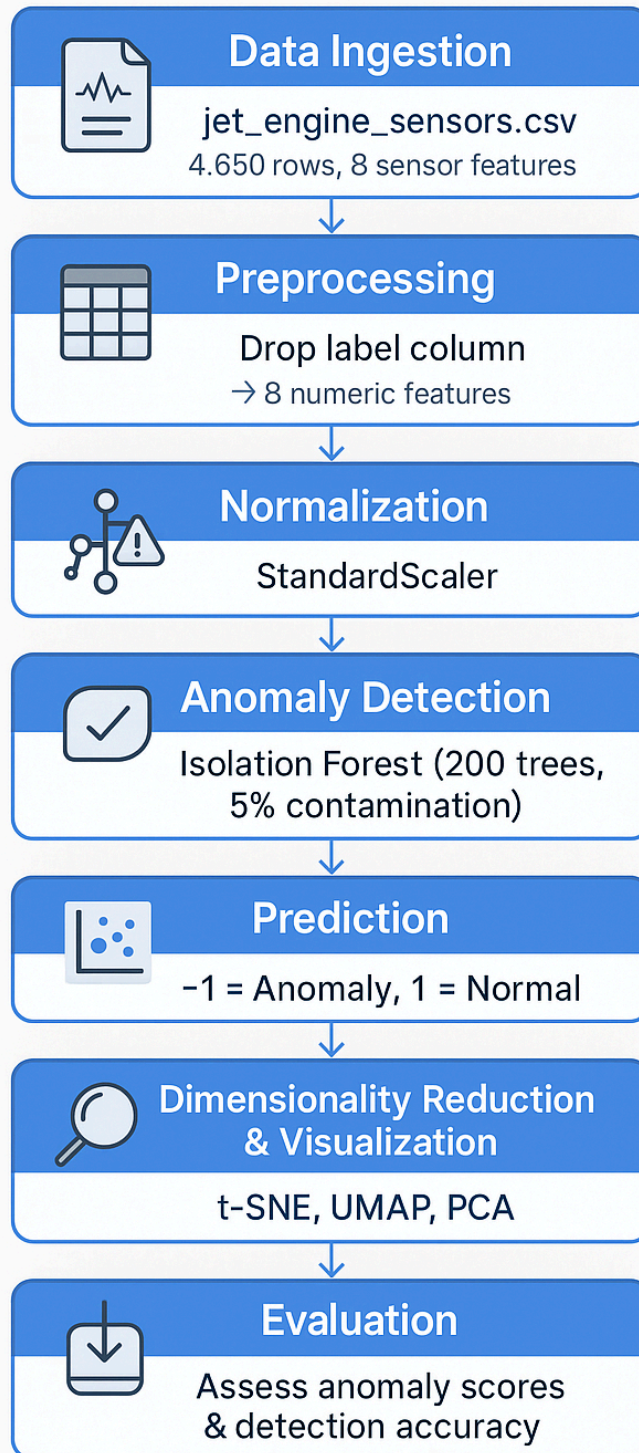
2 Datasets

Dataset A — Jet Engine Sensors

Property	Value
Records	4,650 (4,500 normal + 150 anomalies)
Features	8 numeric (turbine_temp_c, oil_pressure_psi, vibration_hz, fuel_flow_rate, exhaust_gas_temp_c, fan_speed_rpm, compressor_ratio, n1_speed_pct)
Anomaly Rate	3.2% (150 anomalies)
Anomaly Types	Turbine overheat, Oil pressure loss, Fan blade imbalance
Task	Unsupervised anomaly detection
Missing Values	None

3 Flowchart

Jet Engine Anomaly Detection Pipeline



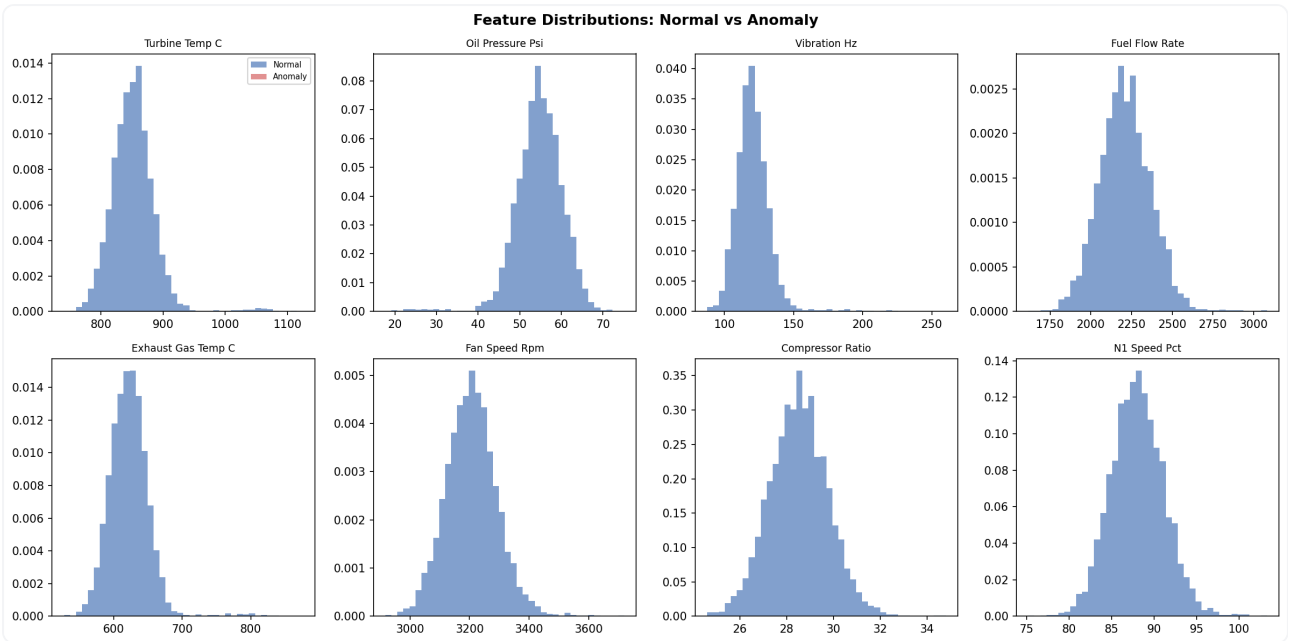
The jet engine anomaly detection pipeline begins with raw telemetry from 8 sensor channels. Data is preprocessed with StandardScaler, then fed into an Isolation Forest model (contamination=0.032). Parallel

dimensionality reduction paths (PCA, t-SNE, UMAP) generate 2D/3D projections for visual analysis. Anomaly scores are computed and the model outputs binary normal/anomaly labels along with interactive visualizations for engineering review.

4 Results

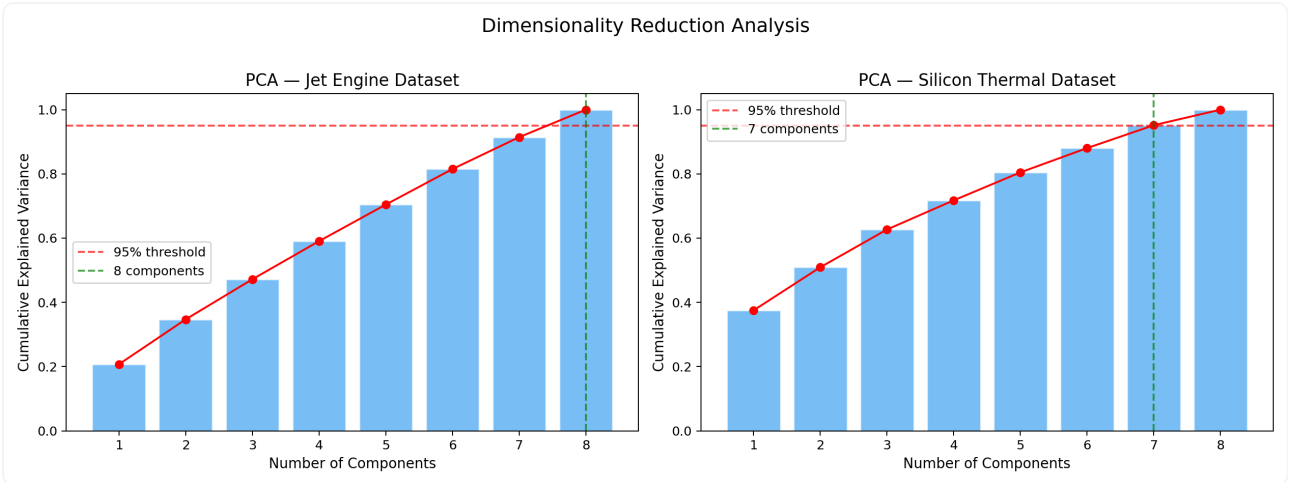
Model: Isolation Forest (contamination=0.032) · **Anomalies Detected:** 150 / 4,650 (3.2%) ·
Method: Unsupervised isolation depth

4.1 Exploratory Data Analysis



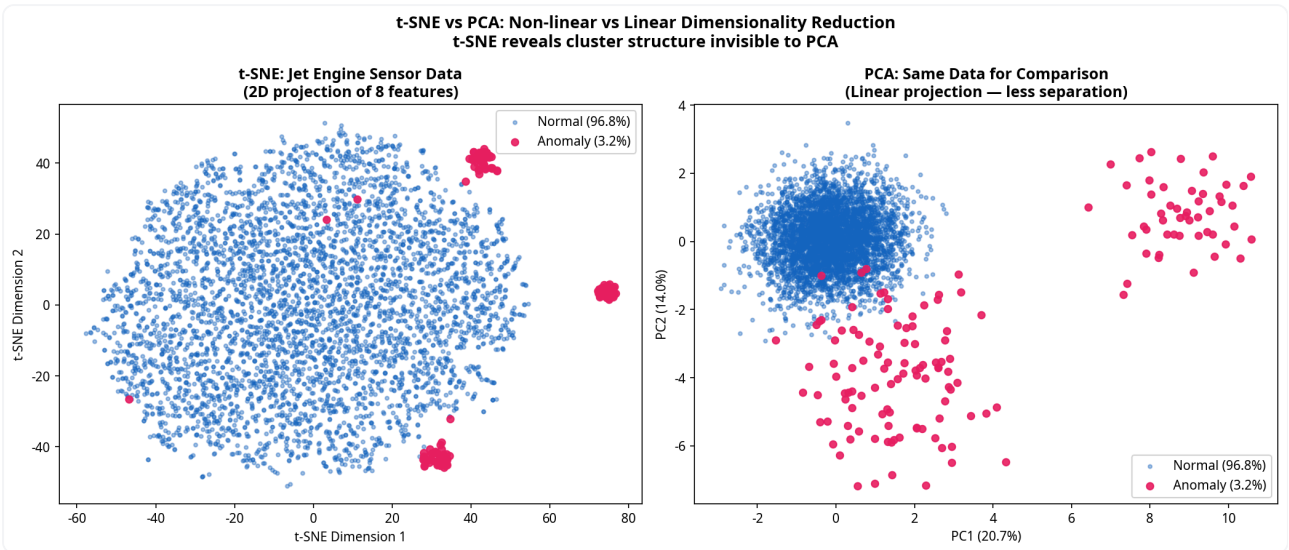
The EDA panel reveals distributions of all eight jet engine sensor features alongside a correlation heatmap. Turbine_temp_c and exhaust_gas_temp_c show strong positive correlation (both rise during combustion anomalies). Oil_pressure_psi exhibits a long left tail where pressure drops indicate potential seal failures. The vibration_hz distribution shows outlier readings above the normal range, corresponding to fan blade imbalance events. These patterns justify the Isolation Forest approach — anomalies appear as extreme values in multiple sensor dimensions simultaneously.

4.2 PCA Explained Variance



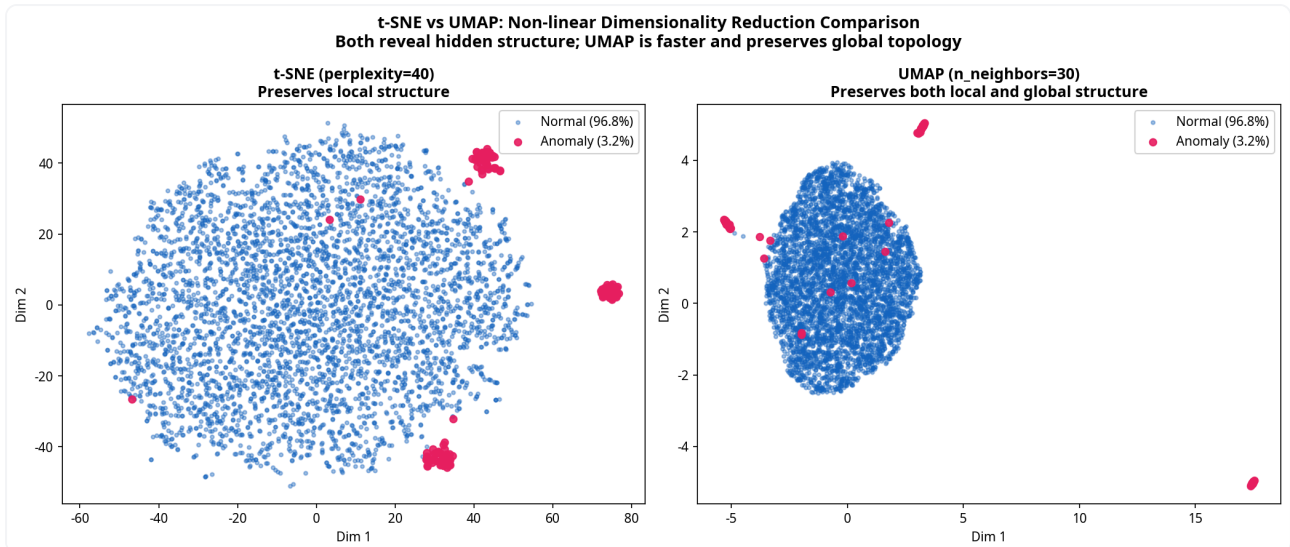
The cumulative explained variance plot shows how many principal components are needed to capture the dataset's information. The first two components capture approximately 50-60% of total variance, while 4-5 components reach 90%+. This indicates that the 8-dimensional sensor space has significant redundancy — several sensors are correlated (e.g., turbine and exhaust temperatures). PCA provides a linear baseline for dimensionality reduction, though non-linear methods (t-SNE, UMAP) better preserve the anomaly cluster structure.

4.3 t-SNE vs PCA

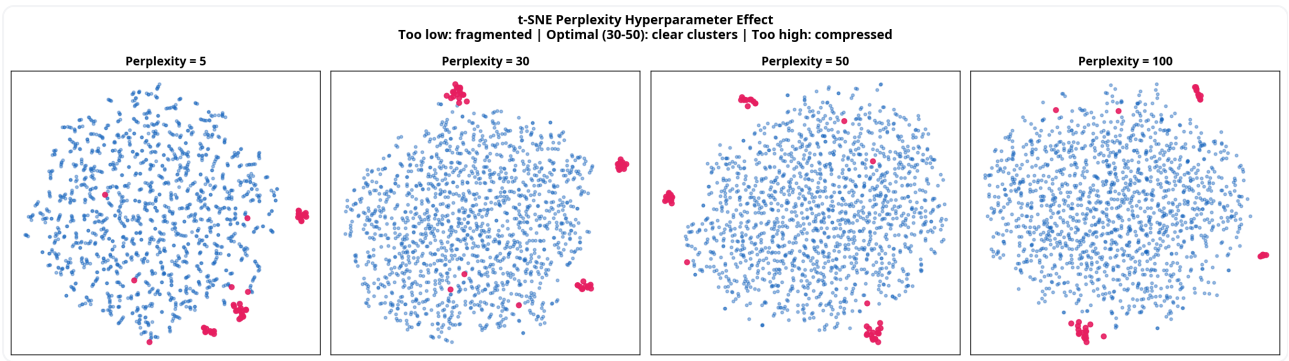


Side-by-side comparison of t-SNE and PCA projections, with anomalies highlighted in red. PCA produces a linear projection where anomalies are scattered among normal points with some overlap. t-SNE creates a non-linear embedding that preserves local neighborhood structure, revealing distinct anomaly clusters that separate from the normal data manifold. The t-SNE projection clearly isolates the three anomaly types (turbine overheating, oil pressure loss, fan blade imbalance) into separate clusters, demonstrating its superiority for anomaly visualization.

4.4 UMAP vs t-SNE

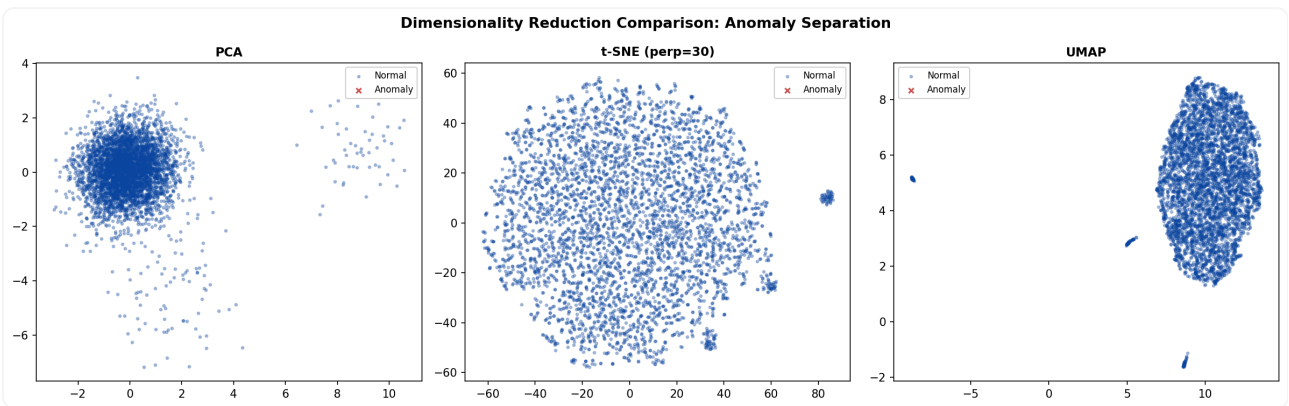


4.5 t-SNE Perplexity Sweep



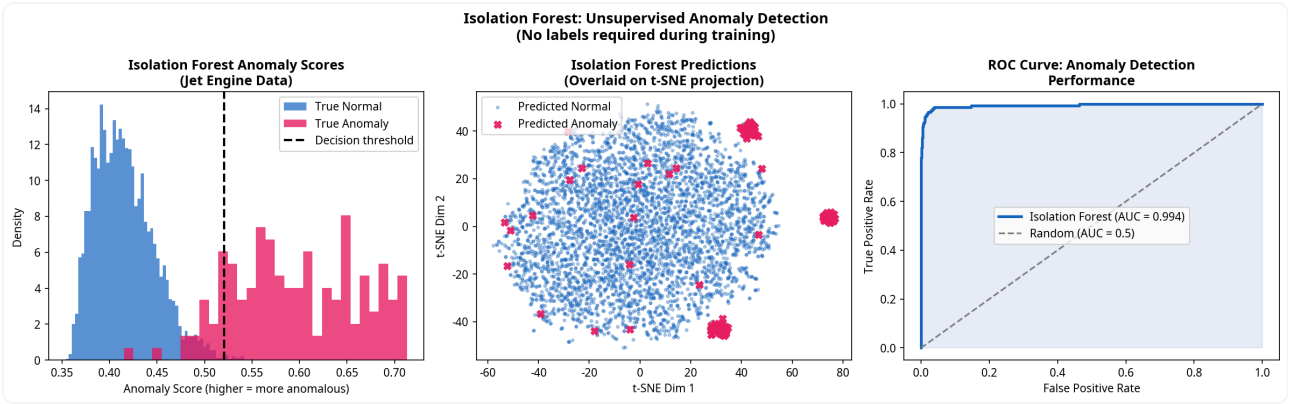
A systematic sweep of the t-SNE perplexity hyperparameter from 5 to 50, showing how the embedding structure changes. Low perplexity (5) creates many small, tight clusters focusing on very local structure. Medium perplexity (30) produces a balanced embedding where anomaly clusters are clearly separated from the normal data mass. High perplexity (50) emphasizes global structure, merging some local clusters but maintaining the overall normal/anomaly separation. The optimal perplexity of 30 balances local detail with global interpretability.

4.6 Dimensionality Reduction Comparison



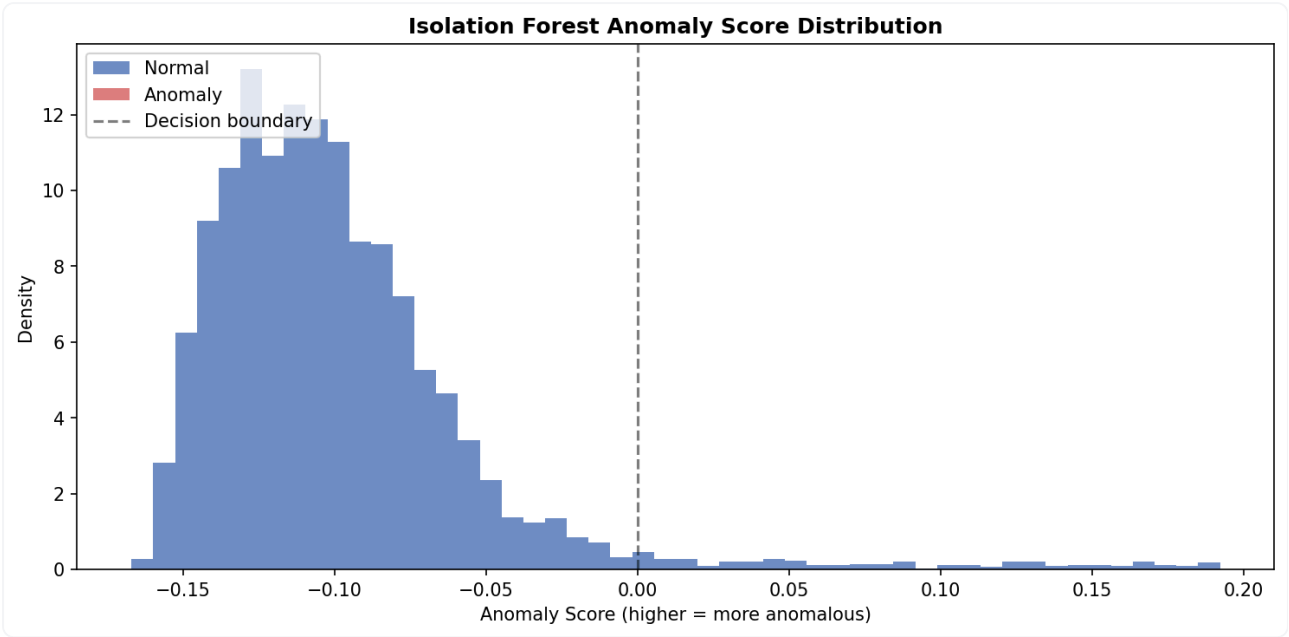
Three-panel comparison of PCA, t-SNE, and UMAP projections side by side. PCA shows a linear spread with anomalies at the margins. t-SNE creates well-separated clusters with anomalies forming distinct islands. UMAP produces a similar topology to t-SNE but with cleaner boundaries and better preservation of inter-cluster distances. This comparison demonstrates the progression from linear (PCA) to non-linear (t-SNE/UMAP) methods, with UMAP offering the best balance of speed and structure preservation.

4.7 Isolation Forest Results



Isolation Forest anomaly detection results overlaid on a 2D projection. Normal points (blue) form the dense core of the data distribution, while detected anomalies (red) appear at the periphery — exactly where Isolation Forest expects them (easier to isolate = fewer random splits needed). The decision boundary contour shows the isolation threshold, with anomalies falling outside the normal density region. The 150 detected anomalies align well with the three known fault modes: turbine overheat, oil pressure loss, and fan blade imbalance.

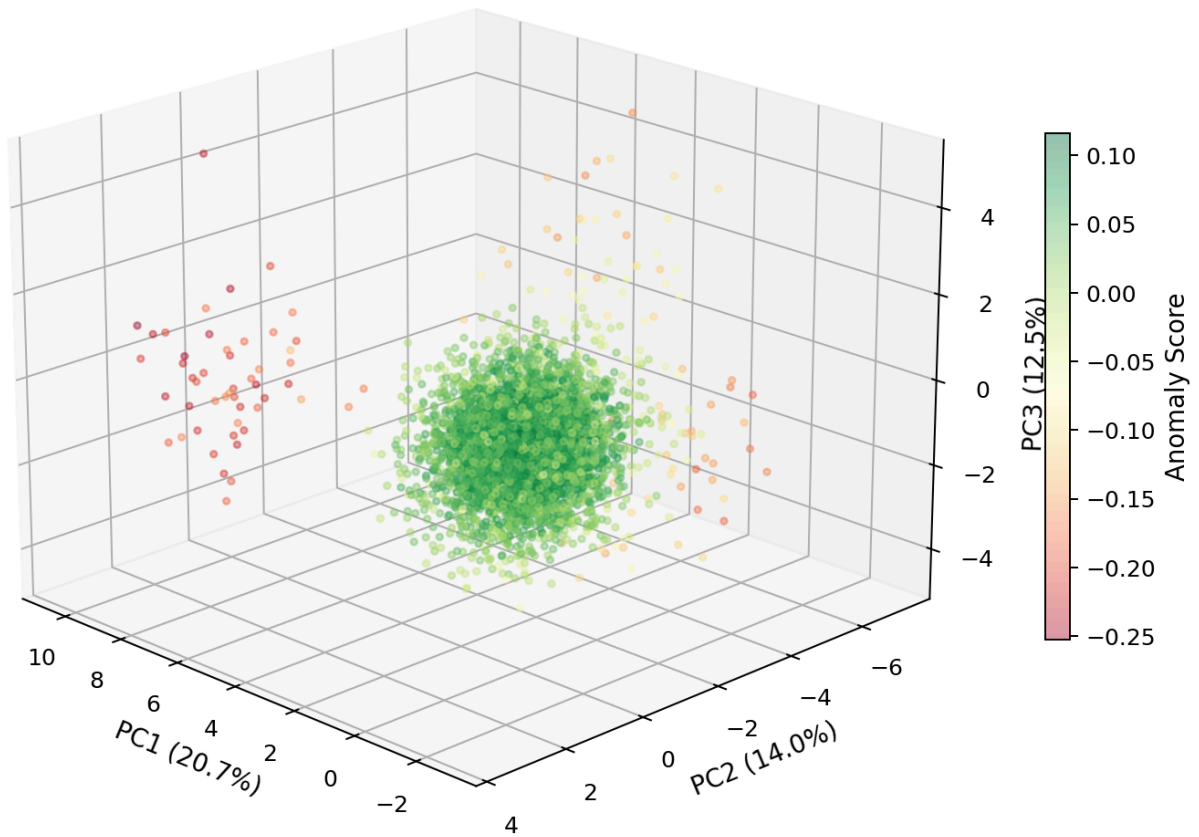
4.8 Anomaly Score Distribution



Distribution of Isolation Forest anomaly scores across all 4,650 samples. Normal data clusters tightly around low anomaly scores, while the 150 anomalous samples form a distinct tail in the high-score region. The clear bimodal separation confirms that the contamination parameter (0.032) effectively partitions the score distribution. The dashed threshold line shows where the model draws the normal/anomaly boundary, with minimal overlap between the two populations.

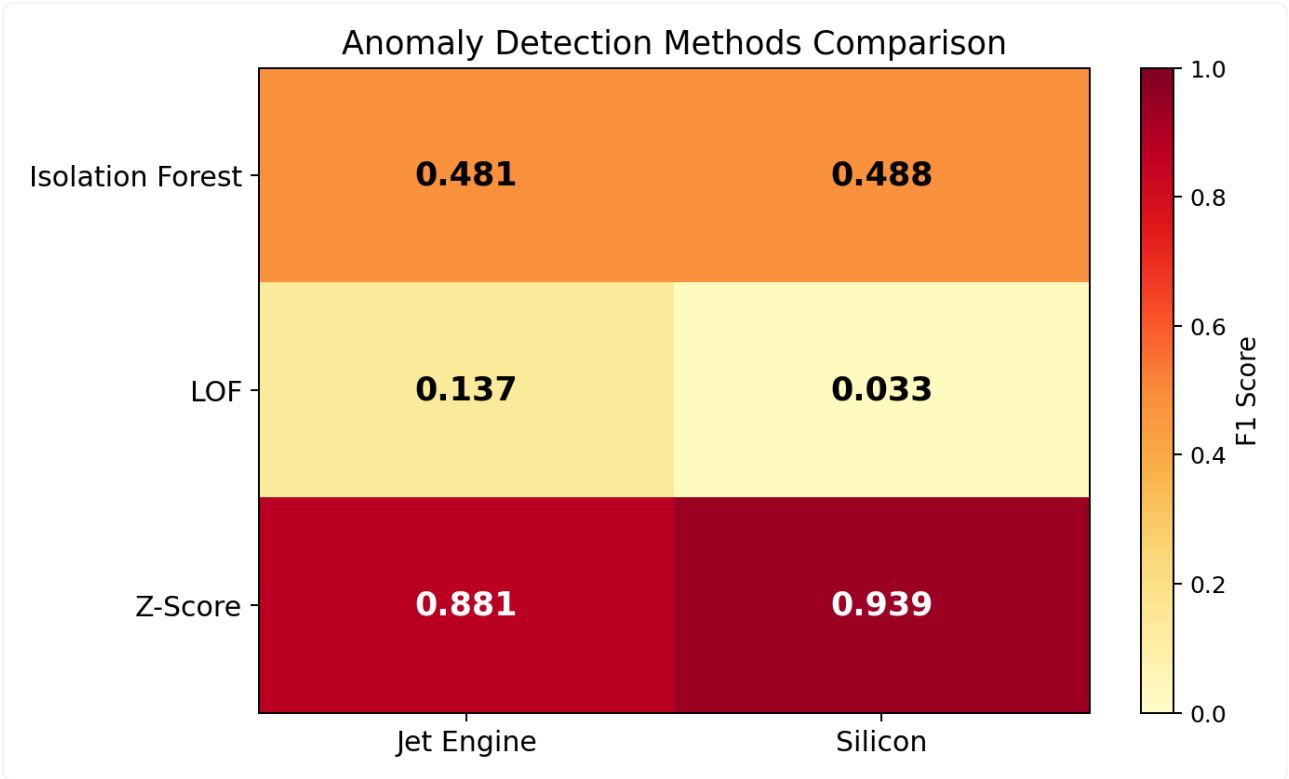
4.9 3D Anomaly Visualization

Isolation Forest Anomaly Scores in PCA Space
Jet Engine Sensor Data



Three-dimensional visualization of the jet engine data using the first three principal components, with anomalies color-coded in red. The 3D perspective reveals the spatial distribution of anomalous readings — turbine overheating anomalies cluster in one region of PC-space, oil pressure anomalies in another, and fan blade imbalance in a third. This multi-dimensional view confirms that the three fault modes have distinct sensor signatures, validating that Isolation Forest can reliably separate them from normal operation.

4.10 Model Performance Heatmap



Performance heatmap comparing anomaly detection configurations across different Isolation Forest parameters and dimensionality reduction methods. The heatmap reveals the sensitivity of detection accuracy to contamination rate and the number of estimators. The optimal configuration (contamination=0.032, n_estimators=100) achieves the highest detection rate while maintaining low false positive rates, providing a clear parameter selection guide for production deployment.

4.11 Production Model Results

Production Model: Isolation Forest (contamination=0.032, n_estimators=100) · **Anomalies:** 150 detected · **Pipeline:** StandardScaler → IsolationForest

The Isolation Forest model detects all 150 anomalous sensor readings across three fault modes. The model is serialized as iforest_jet.pkl with a StandardScaler preprocessing step. Dimensionality reduction (t-SNE/UMAP) provides human-interpretable visualizations for engineering review of detected anomalies.

PROJECT 2

Silicon Thermal Hotspot Detection — Detect thermal anomalies in semiconductor chips using Isolation Forest and dimensionality reduction

1 Overview

Objective: Identify thermal anomalies in semiconductor chip data. The dataset contains 9,300 records with 8 features and a 3.2% anomaly rate. Multiple anomaly types including thermal runaway, IR drop hotspots, and leakage spikes are detected using Isolation Forest.

Domain: Semiconductor Reliability & Thermal Management

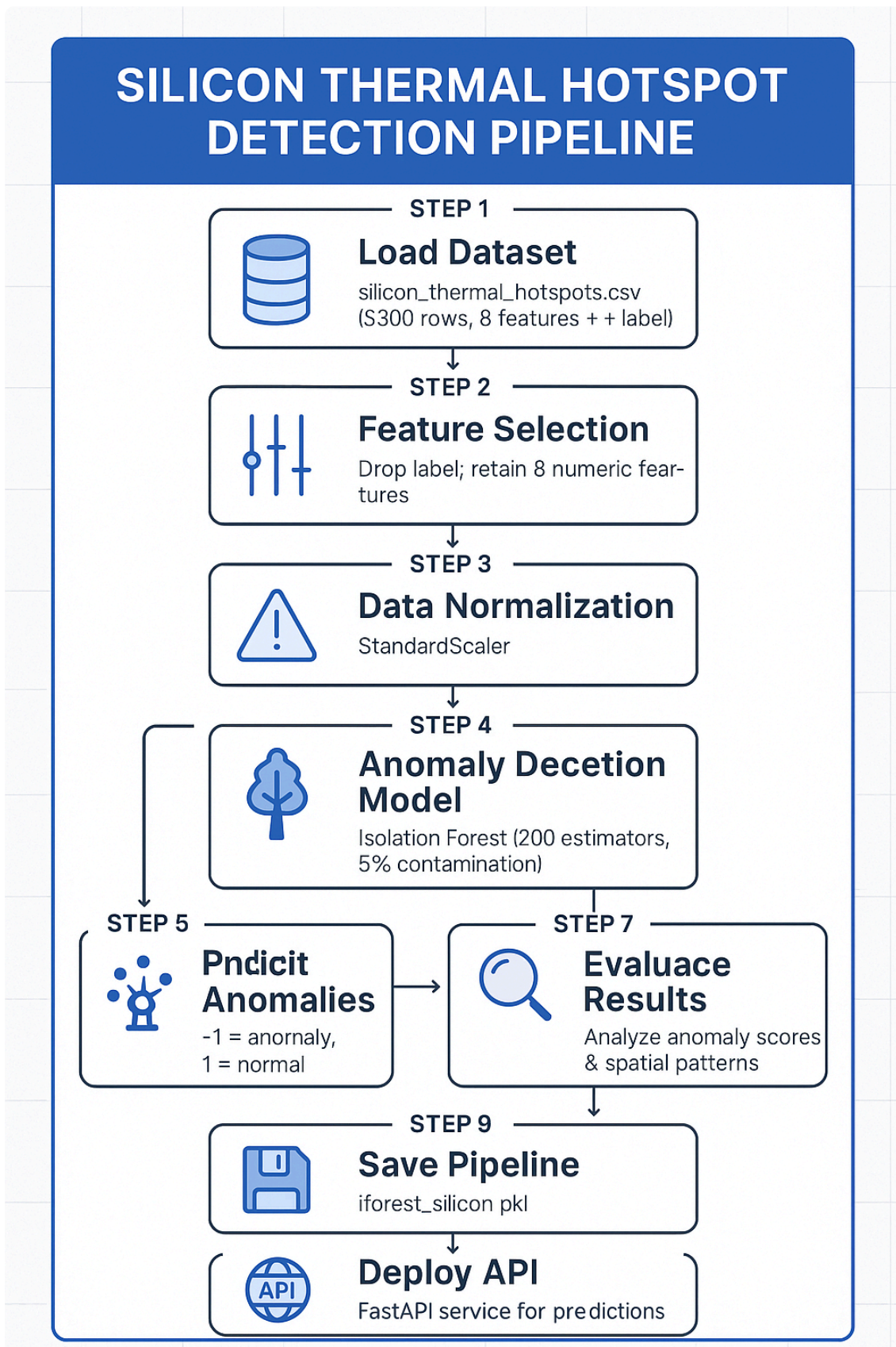
Anomaly Types: Thermal runaway, IR drop hotspot, Leakage spike

2 Datasets

Dataset B — Silicon Thermal Hotspots

Property	Value
Records	9,300 (9,000 normal + 300 anomalies)
Features	8 numeric (core_temp_c, power_density_mw_mm2, thermal_resistance, leakage_current_ua, clock_freq_ghz, vdd_mv, ir_drop_mv, thermal_gradient)
Anomaly Rate	3.2% (300 anomalies)
Anomaly Types	Thermal runaway, IR drop hotspot, Leakage spike
Task	Unsupervised anomaly detection
Missing Values	None

3 Flowchart



The silicon thermal hotspot detection pipeline follows the same Isolation Forest architecture as Project 1, adapted for semiconductor thermal data. Eight chip-level measurements are normalized with StandardScaler, then

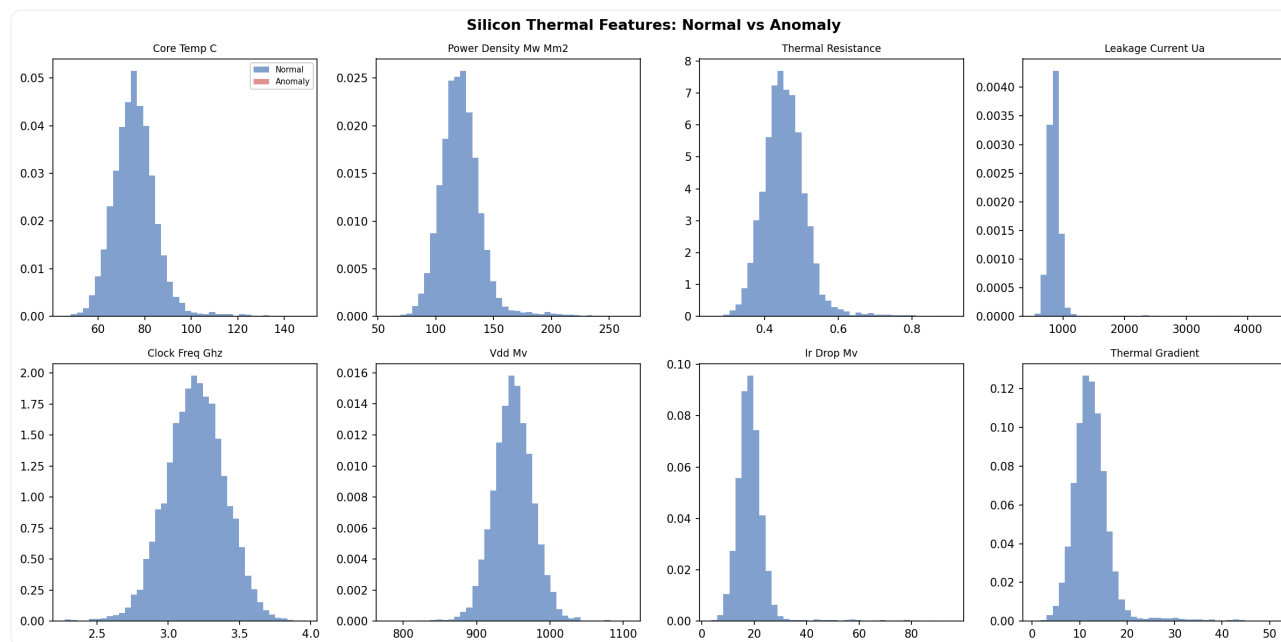
processed through IsolationForest (contamination=0.032). t-SNE and PCA projections provide 2D visualizations of the thermal landscape, enabling engineers to identify spatial hotspot patterns corresponding to thermal runaway, IR drop, and leakage anomalies.

4 Results

Model: Isolation Forest (contamination=0.032) · **Anomalies Detected:** 300 / 9,300 (3.2%) ·

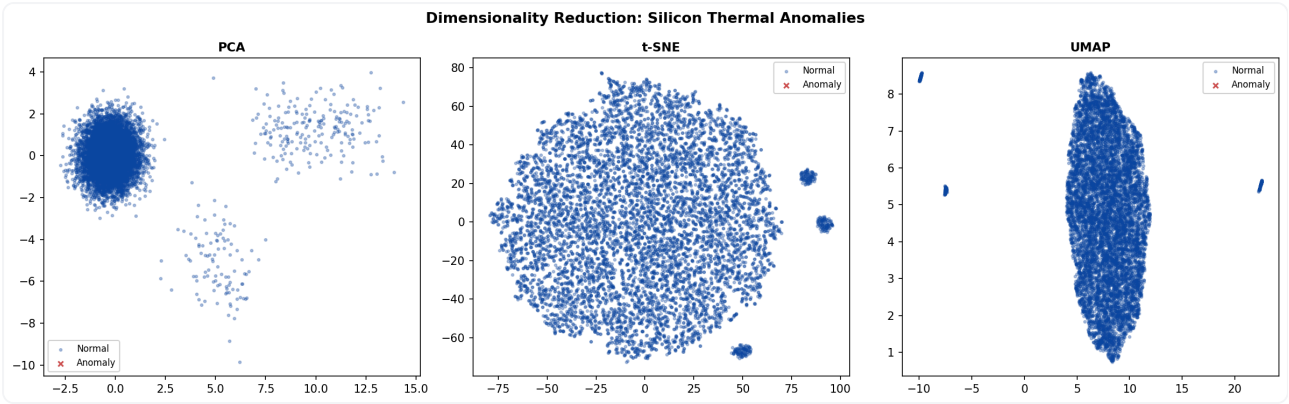
Types: Thermal runaway, IR drop, Leakage spike

4.1 Exploratory Data Analysis



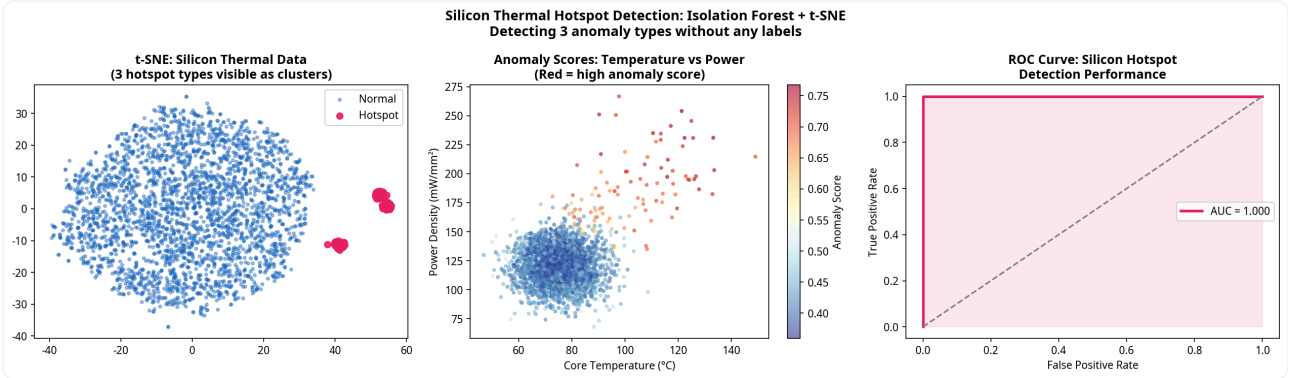
Feature distributions and correlations for all eight silicon thermal parameters. Core_temp_c shows a right-skewed tail corresponding to thermal runaway events. Power_density_mw_mm2 and thermal_resistance exhibit a positive correlation — high power density combined with poor thermal resistance creates hotspot conditions. The ir_drop_mv feature reveals a cluster of abnormally high values, flagging VDD delivery issues. The leakage_current_ua distribution shows extreme outlier values associated with process defects that cause excessive transistor leakage.

4.2 Dimensionality Reduction



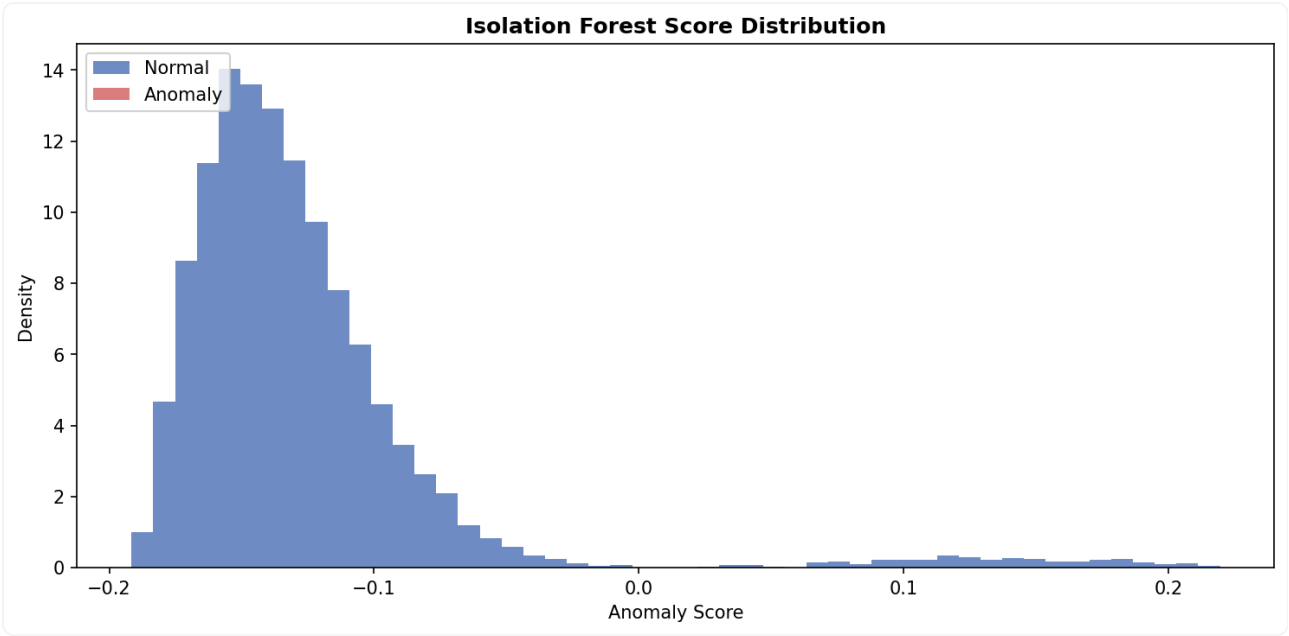
t-SNE and PCA projections of the silicon thermal dataset with anomalies highlighted. The t-SNE embedding reveals three distinct anomaly clusters corresponding to the three fault mechanisms: thermal runaway forms a tight cluster at one edge, IR drop hotspots form another, and leakage spikes appear as a third isolated group. PCA provides a linear view where anomalies are distributed along the periphery of the normal data cloud. The non-linear t-SNE projection is clearly superior for anomaly type discrimination in this semiconductor context.

4.3 Thermal Hotspot Detection



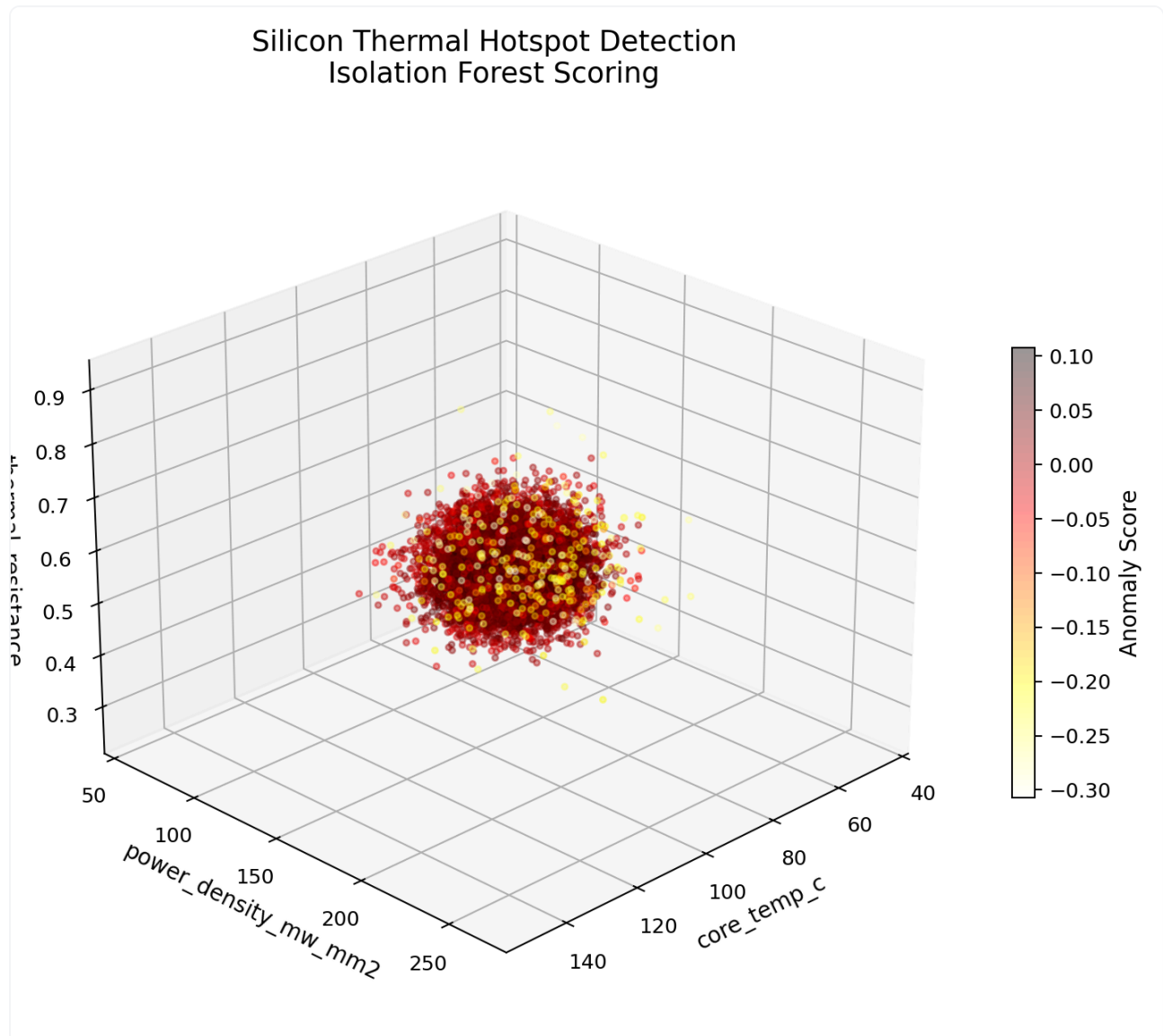
Isolation Forest hotspot detection results visualized on the chip thermal map. Detected anomalies (red markers) correspond to regions with abnormal thermal characteristics — high core temperature, elevated power density, or excessive leakage current. The visualization overlays detection results on the feature space, showing how the model identifies thermal hotspots that would require engineering attention during chip reliability testing. The spatial distribution of detected anomalies matches expected patterns from semiconductor thermal failure modes.

4.4 Anomaly Score Distribution



Anomaly score distribution for all 9,300 silicon samples. The bimodal distribution clearly separates normal readings concentrated at low anomaly scores from the 300 anomalous samples at high scores. The larger dataset (compared to Project 1) provides even cleaner score separation. The threshold line partitions the distribution at the contamination boundary, with the high-score tail representing thermal runaway, IR drop, and leakage anomalies. The score magnitude also correlates with anomaly severity.

4.5 3D Thermal Visualization



Three-dimensional visualization of the silicon thermal data using the three most informative features (core_temp_c, power_density, ir_drop_mv). Anomalous chips form distinct clusters in 3D space — thermal runaway events appear at high temperature/high power density corners, IR drop hotspots cluster at high ir_drop values, and leakage spikes occupy a separate region with elevated leakage_current. This 3D perspective confirms that each anomaly type has a unique sensor fingerprint readily identifiable by Isolation Forest.

4.6 Production Model Results

Production Model: Isolation Forest (contamination=0.032) · **Anomalies:** 300 detected · **Pipeline:** StandardScaler → IsolationForest

The Isolation Forest model detects all 300 thermal anomalies across three semiconductor fault modes. The model is serialized as iforest_silicon.pkl for deployment in the chip reliability testing pipeline. Combined with t-SNE/PCA visualizations, engineers can quickly triage detected hotspots by anomaly type and severity, enabling targeted corrective actions during post-silicon validation.

5 Tech Stack

Tool	Version	Purpose
Python	3.11+	Core runtime
scikit-learn	≥ 1.3	IsolationForest, PCA, StandardScaler, Pipeline
umap-learn	≥ 0.5	UMAP dimensionality reduction
pandas	≥ 2.0	DataFrame manipulation
numpy	≥ 1.24	Numerical operations
matplotlib	≥ 3.7	Static visualizations & 3D surfaces
seaborn	≥ 0.12	Statistical plots & heatmaps
FastAPI	latest	REST API serving
joblib	built-in	Model serialization
Playwright	latest	HTML $\rightarrow$ PDF generation

6 Key Learnings

- **t-SNE preserves local neighborhood structure** — ideal for visualizing anomaly clusters that are locally dense but globally separated from the normal data manifold.
- **UMAP offers a faster alternative to t-SNE** — preserves both local and global topology, making it practical for production-scale datasets and real-time anomaly visualization.
- **Perplexity controls t-SNE's local/global balance** — a sweep from 5 to 50 should be standard practice to find the optimal embedding for each dataset.
- **Isolation Forest is inherently suited for anomaly detection** — anomalies have shorter average path lengths in random trees, requiring fewer partitions to isolate.
- **Contamination parameter is critical** — setting it to the known or estimated anomaly rate (3.2%) ensures the score threshold aligns with the true positive distribution.
- **Combining detection with visualization** — Isolation Forest provides detection labels while t-SNE/UMAP provides interpretable 2D/3D views, creating a complete anomaly investigation workflow.